FlashKDA and KDA Kernels

Deep Dive into Kimi K3 — Chapter 34

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Goal: fix the semantic contract, derive the affine segment interface, trace FlashKDA's K1/K2 split through one packed call, carry it into backward, context parallelism, and decode recovery — and keep every timing tied to its measured setup.

Roadmap

The contract and its boundaries

Affine segments and the K1/K2 split

Backward, context parallelism, and decode

Costs and evidence scope

The contract and its boundaries

Same boundaries, whatever the schedule

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Matching current outputs cannot certify continuation: the next call consumes the state.

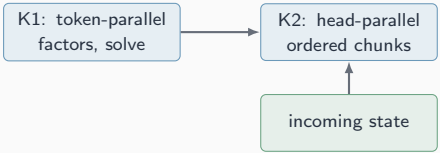
Affine segments and the K1/K2 split

Prepare locally, advance in order

K1: token-parallel factors, solve

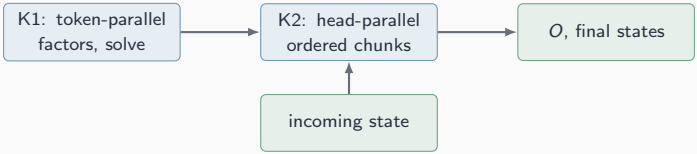
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Few heads per GPU leave K2 too few chains — the case for context parallelism.

Backward, context parallelism, and decode

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Public FlashKDA is forward-only; gradients and CP use generic FLA; K3's internal kernels remain reported, not inspected.

Costs and evidence scope

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Next, Chapter 35 applies the same discipline to hybrid-attention and depth-mixing kernels.